

Proton Beam Therapy of Parapapillary Choroidal Melanoma

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- **PURPOSE:** To analyze the functional outcome with regard to the development of visual acuity and radiation-induced optic neuropathy of patients with parapapillary choroidal melanoma treated with proton beam therapy.
- **DESIGN:** Clinical case series, retrospective study.
- **METHODS:** We evaluated 147 consecutive patients with parapapillary choroidal melanoma who received proton beam therapy as primary tumor treatment at the Helmholtz Center in Berlin from 1998 to 2005. A cumulative dose of 60 Cobalt Gray Equivalents (CGE) was delivered to the tumor and the optic disc received a minimum of 50 CGE. Kaplan-Meier analysis was used to assess ocular outcome and survival rates. For trend analysis of functional development, Wilcoxon-Mann-Whitney *U* test was used to compare the medians of 2 groups and Kruskal-Wallis test was used in the case of more than 2 groups.
- **RESULTS:** The mean follow-up time was 6.5 years (range 0.3-11.7 years). The most common side effects were radiation-induced optic neuropathy, retinopathy, and cataract. The median visual acuity before and within the first year after therapy was 0.4 logMAR (20/50), lapsing to 1.3 logMAR (20/400) after 3 years and 1.4 logMAR (20/500) after 5 years. During follow-up, no light perception developed in 17 cases (11.6%), mostly attributed to radiation-induced retinopathy, optic neuropathy, and secondary glaucoma. Enucleation was carried out in 14 patients (9.5%) because of local recurrence or severe side effects.
- **CONCLUSION:** Radiation-induced optic neuropathy is an expected issue after proton beam therapy of parapapillary choroidal melanoma, and visual impairment is

common during long-term follow-up, but some useful vision can be preserved in a considerable number of patients. (*Am J Ophthalmol* 2014;157:1258-1265. © 2014 by Elsevier Inc. All rights reserved.)

THE COLLABORATIVE OCULAR MELANOMA STUDY (COMS)¹ indicated that iodine plaque irradiation of medium-sized choroidal melanoma did not influence the overall survival rate as compared to primary enucleation. Today, irradiation has become widely accepted as a globe-preserving treatment modality in the majority of choroidal melanomas. Irradiation techniques comprise brachytherapy with, for example, iodine-125 or ruthenium-106 plaques or teletherapy; foremost proton beam therapy; and stereotactic radiosurgery (gamma-knife). In tumors with para- or peripapillary locations, the involvement of the optic disc is unavoidable, since it has to be included in the treatment target irrespective of the chosen treatment technique. Irradiation induces microvasculopathy owing to endothelial cell damage of the small vessels, leading to vascular occlusion,² edema, and atrophy of the optic disc in the late stage. Radiation-induced optic neuropathy is anticipated in these cases. Parapapillary location is a relative constraint for brachytherapy, as a sufficient irradiation warrants plaque placement beneath the tumor, which might be hindered by the optic nerve. Therefore, these tumors have been excluded from the COMS study.

Though notched plaques allow the treatment of juxtapapillary melanomas, the correct placement is complex and prone to malpositioning or tilting.³ Especially in irregularly shaped tumors or those with a large base, local tumor control can only be ensured at the expense of damaging the surrounding normal tissue, since it has to be included in the irradiation field. Moreover, brachytherapy affects dose distributions that typically account for very high radiation loads delivered to the macula and optic disc when the plaque has to be sutured close to these critical structures. Therefore, eccentric plaque placement has been practiced in some centers, taking into consideration the possibility of a higher recurrence rate.⁴ Whereas brachytherapy is a rather common method, there are few centers worldwide that offer proton beam therapy.⁵⁻⁹ The advantage of proton beam therapy is the individually matched beam, offering the possibility of sparing critical structures of the

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